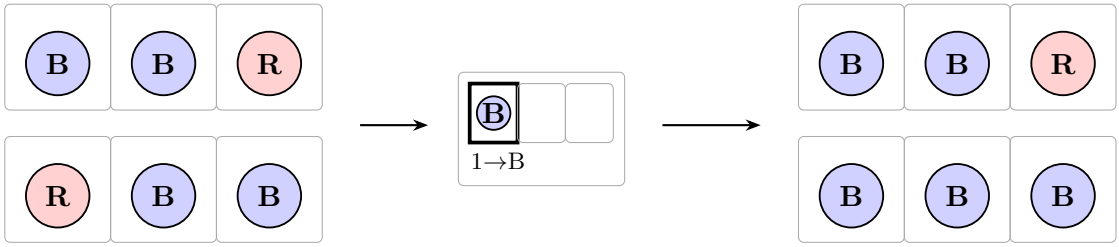
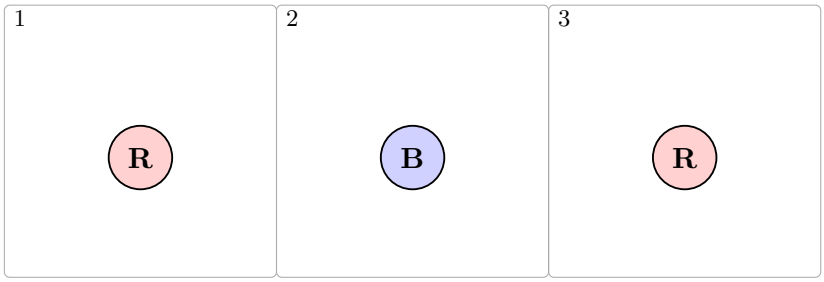
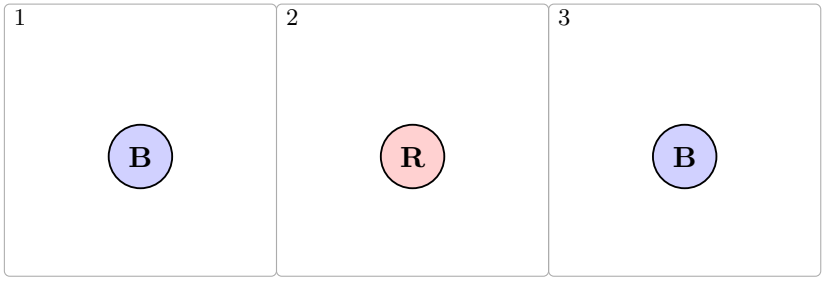
Each instruction names a slot and a color. Set that slot to that color on both boards, even if it is already that color. Slots are 1, 2, 3 from left to right.

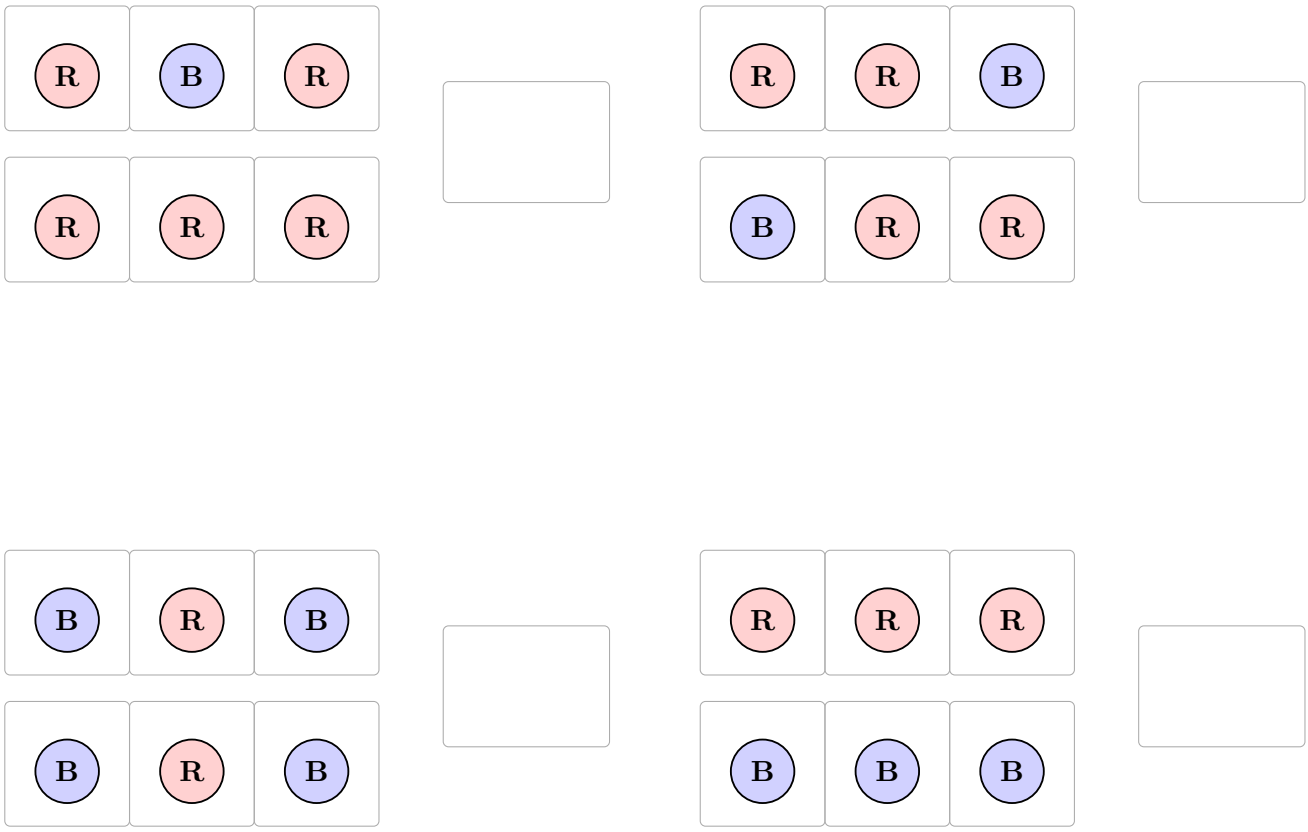


Problem 1: Make these boards match using as few instructions as you can. Can any other starts need more instructions?



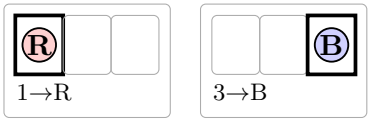


Problem 2: How few instructions can make each pair match? Try every pair.



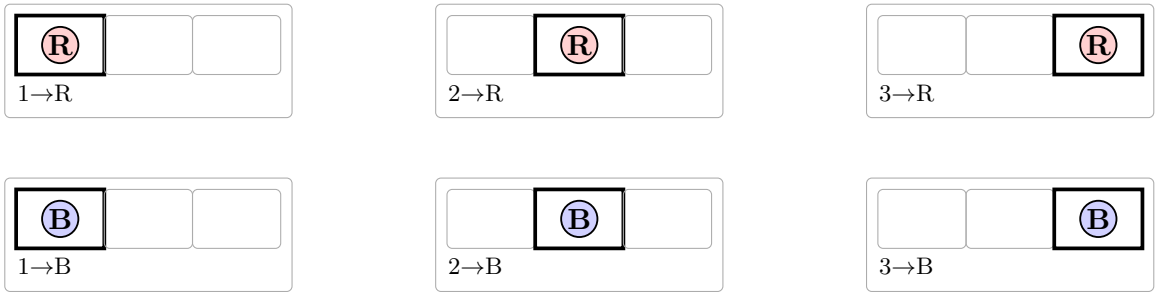
Problem 3: Invent a pair that needs exactly two instructions to match. Can you invent a pair that needs four?

Problem 4: Use these two instructions. Make two different starts that end matching, and a pair that ends different.

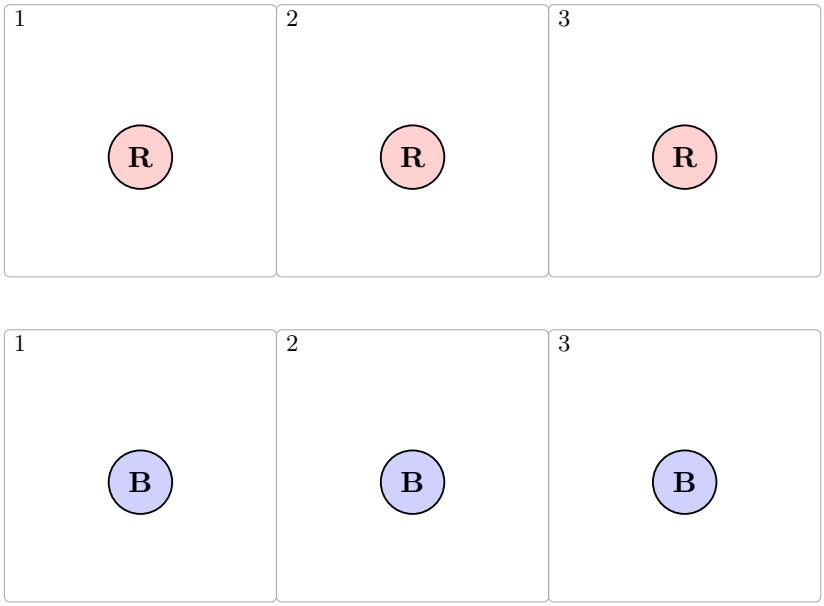


Problem 5: Choose three instructions that make any two starts match. Could just two instructions always do it?

Mix these six cards in a cup. Draw one, use it on both boards, then put it back and mix.



Problem 6: Try six draws from these starts, then start again. Could matching take more than three draws?



Problem 7: Keep drawing after the boards match. Can any instruction make them different again?